Breast Cancer Classification with Machine Learning

A Comparative Analysis of Multiple Classification Models Vector Machines

Alessia Corica

MSc in Bioinformatics, University of Bologna

Abstract

Motivation: Breast cancer is one of the most common cancers worldwide, and detecting it early is essential to improve patient outcomes. Machine learning can help in this context by finding patterns in clinical data that are not immediately visible through traditional analysis, making it easier to distinguish between benign and malignant cases.

Methods: A machine learning pipeline was developed to classify breast cancer cases using the Breast Cancer Wisconsin Diagnostic dataset. The workflow included exploratory data analysis, feature scaling, and correlation analysis. A stratified train/test split was applied, and multiple classifiers were trained, including Logistic Regression, Support Vector Machines (SVM), Linear Discriminant Analysis (LDA), and K-Nearest Neighbors (KNN).

Results: All models performed well on the test set. SVM showed the best overall performance, with the highest F1-score (0.992) and high ROC-AUC values, while Logistic Regression also achieved strong and well-balanced results (F1 = 0.968). KNN performed well, although with slightly lower recall, indicating a higher tendency to miss some malignant cases. LDA showed the lowest performance, particularly in terms of recall (0.875). Overall, the main differences between the models concern their ability to correctly identify malignant cases.

Contact: alessia.corica@studio.unibo.it

Supplementary material: GitHub repository

Contents

1 Introduction **3**

1.1 Biological background . . . 3

1.2 Machine Learning for Medical Diagnosis . . . 3

1.3 Classification Models . . . 3

1.4 Aim of the Study . . . 4

2 Materials and Methods **4**

2.1 Dataset Overview . . . 4

2.2 Dataset Analysis . . . 5

2.3 Preprocessing . . . 5

2.4 Model Development . . . 5

2.5 Cross-Validation . . . 6

2.6 Model Evaluation . . . 6

3 Results **7**

3.1 Exploratory Data Analysis . . . 7

3.2 Overall Classification Performance . . . 8

3.3 ROC analysis . . . 9

3.4 Precision–Recall analysis . . . 9

3.5 Error Analysis and Confusion Matrices . . . 10

4 Discussion **11**

5 Conclusion **12**

1 Introduction

1.1 Biological background

Breast cancer is one of the most common cancers worldwide and represents a major public health issue. Its development is influenced by a combination of genetic, environmental, and lifestyle factors, which together contribute to tumor onset and progression. Although screening and treatment have improved over time, early detection is still a key factor in increasing patient survival and improving clinical outcomes.

In recent years, technological advances have greatly improved diagnostic capabilities. In particular, high-throughput sequencing and machine learning (ML) have introduced new opportunities for cancer detection and risk assessment. These approaches highlight the increasing role of computational methods in supporting medical decision-making and improving diagnostic accuracy.

1.2 Machine Learning for Medical Diagnosis

ML is increasingly being applied in medical diagnosis to support clinical decision-making. Diagnosing diseases can be challenging because symptoms are often ambiguous and require interpretation by experienced healthcare professionals. In addition, traditional diagnostic procedures may be time-consuming, expensive, and prone to human error.

ML approaches provide therefore data-driven methods that can assist clinicians by identifying patterns in complex medical data. These models can analyze various types of healthcare information, including medical images, clinical records, and patient attributes such as age or medical history. As a result, ML-based diagnostic systems have the potential to improve diagnostic accuracy, reduce costs and accelerate the detection of diseases in clinical practice.

1.3 Classification Models

In this study, multiple supervised ML algorithms were employed to perform binary classification of breast cancer cases, including Logistic Regression, Support Vector Machines (SVM), Linear Discriminant Analysis (LDA), and K-Nearest Neighbors (KNN).

Logistic Regression is a widely used baseline model that estimates class probabilities using a sigmoid function applied to a linear combination of input features. Its simplicity and interpretability make it particularly suitable for biomedical applications.

SVM aim to find the optimal separating hyperplane between classes by maximizing the margin between them. Through the use of kernel functions, such as the radial basis function (RBF), SVMs can model non-linear decision boundaries and capture complex patterns in the data.

LDA is a statistical approach that projects the data into a lower-dimensional space while maximizing class separability. It assumes that the data for each class follow a Gaussian distribution and share the same covariance, making it efficient and effective when these assumptions are approximately satisfied.

KNN is a non-parametric method that classifies new observations based on the majority class among their nearest neighbors in the feature space. Its performance depends strongly on the choice of distance metric and the value of k.

1.4 Aim of the Study

The aim of this study is to develop and evaluate a ML pipeline for the classification of breast cancer cases using structured clinical data. The analysis focuses on building predictive models capable of distinguishing between benign and malignant tumors.

The pipeline includes data exploration, preprocessing, model development, and performance evaluation. Multiple classification algorithms are compared to assess their predictive performance and suitability for this task.

Model performance is evaluated using standard classification metrics, including accuracy, precision, recall, F1-score, ROC-AUC, and Precision–Recall curves.

The final goal is to identify the most effective approach for breast cancer classification and to analyze the strengths and limitations of different machine learning methods.

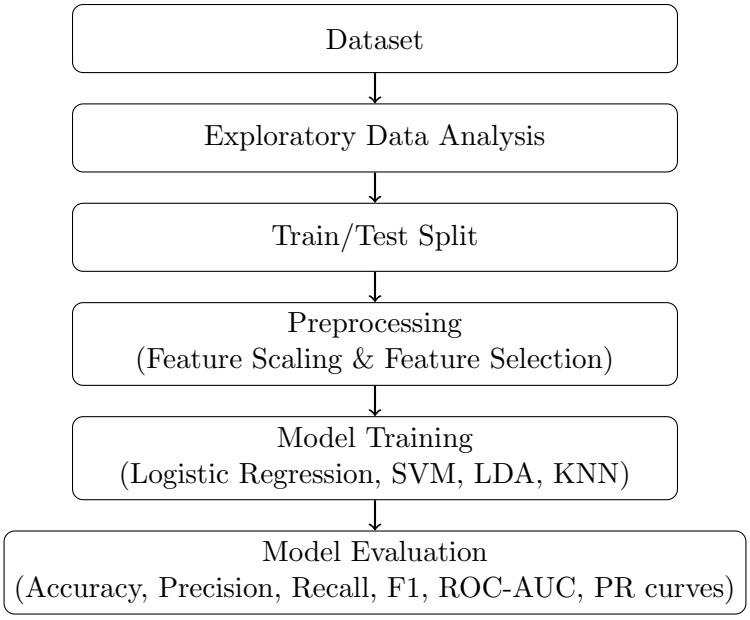


Figure 1: Overview of the ML workflow implemented in this study.

2 Materials and Methods

2.1 Dataset Overview

The dataset used in this study is the *Breast Cancer Wisconsin Diagnostic Dataset*, a widely used benchmark dataset for evaluating classification algorithms in biomedical applications. The data originate from digitized images of fine needle aspirate (FNA) of breast masses, from which several quantitative measurements describing the morphology of cell nuclei are extracted.

Each sample corresponds to a tumor observation and is represented by 30 numerical features that capture geometric and structural properties of the cell nuclei, including radius, texture, perimeter, area, smoothness, compactness, concavity, and symmetry. These measurements provide a quantitative description of tumor morphology and enable the development of predictive models capable of distinguishing between benign and malignant tumors.

Characteristic	Value
Number of samples	569
Number of features	30
Feature type	Numerical
Number of classes	2
Class labels	Benign, Malignant

Table 1: Main characteristics of the Breast Cancer Wisconsin Diagnostic Dataset.

2.2 Dataset Analysis

Before model development, an exploratory analysis of the dataset was performed to assess its suitability for the classification task. The dataset was first inspected for missing or inconsistent values in order to determine whether data cleaning or imputation procedures were required.

The distribution of the diagnostic classes was then examined to evaluate the presence of class imbalance between benign and malignant samples. This step was important to guide the choice of appropriate evaluation metrics and to motivate the use of stratified sampling during the train/test split.

Finally, the numerical predictors were analyzed to investigate their relationships and potential redundancy. In particular, a Pearson correlation matrix was computed across all numerical features to assess dependencies among variables and to support the interpretation of the feature space before model training.

2.3 Preprocessing

Class labels were converted into numerical format (0 for benign and 1 for malignant) to allow their use in machine learning models. The dataset was divided into training and test sets using `train_test_split()` (70% training-30% test) with `random_state=42` to ensure reproducibility and stratification to preserve class proportions across both subset.

During model development, preprocessing steps including feature standardization and feature selection were incorporated directly into the modeling pipeline. Standardization was performed using `StandardScaler` inside cross-validation folds to ensure that all features contribute equally to the models. Feature selection was applied using `SelectKBest` with an ANOVA F-test to reduce redundancy and retain the most informative predictors.

To address class imbalance, the `class_weight='balanced'` option was used in Logistic Regression and SVM models.

2.4 Model Development

Multiple supervised machine learning models were implemented for the classification task: Logistic Regression, SVM, LDA and KNN. These models were selected to compare different learning approaches, including linear and non-linear methods, as well as parametric and non-parametric techniques.

Logistic Regression was used as an interpretable baseline model, while SVM with a radial basis function (RBF) kernel was included to capture non-linear relationships in the data. LDA was considered as a statistical linear classifier based on class separability, and KNN as a non-parametric, distance-based method.

All models were trained within a pipeline that includes feature standardization and fea-

ture selection. These preprocessing steps were applied during cross-validation to ensure a fair comparison across models. Hyperparameter tuning was performed for all classifiers using `GridSearchCV` combined with cross-validation, allowing the selection of optimal model configurations and a more robust estimation of performance.

2.5 Cross-Validation

A stratified 5-fold cross-validation approach was used on the training dataset to obtain a reliable estimate of model performance, ensuring that each fold preserved the original class distribution.

Cross-validation was integrated into the model development process through `GridSearchCV`, allowing for consistent hyperparameter tuning and feature selection across all folds.

2.6 Model Evaluation

The predictive performance of the models was evaluated on the independent test dataset. Several evaluation metrics were computed in order to obtain a comprehensive assessment of classification performance.

- **Accuracy**: proportion of correctly classified observations.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- **Precision**: proportion of predicted positive observations that are actually positive.

$$Precision = \frac{TP}{TP + FP}$$

- **Recall (Sensitivity)**: proportion of actual positive observations correctly identified by the model.

$$Recall = \frac{TP}{TP + FN}$$

- **F1-score**: harmonic mean of precision and recall.

$$F1 = 2 \cdot \frac{Precision \cdot Recall}{Precision + Recall}$$

- **ROC-AUC**: measures the ability of the classifier to discriminate between classes across all possible decision thresholds. The ROC curve is obtained by plotting the True Positive Rate (TPR) against the False Positive Rate (FPR), defined as

$$TPR = \frac{TP}{TP + FN}$$

$$FPR = \frac{FP}{FP + TN}$$

- **PR-AUC (Average Precision)**: evaluates the trade-off between precision and recall across different classification thresholds. It is particularly useful for imbalanced datasets.

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Finally, all classification models were compared using ROC and Precision–Recall curves computed from predicted probabilities (or decision scores), providing a threshold-independent evaluation of models performance.

3 Results

3.1 Exploratory Data Analysis

The class distribution of the dataset shows that benign cases represent the majority of the samples, while malignant cases account for a smaller proportion.

This moderate class imbalance highlights the importance of considering evaluation metrics beyond accuracy, such as recall and F1-score, to properly assess model performance on the minority class.

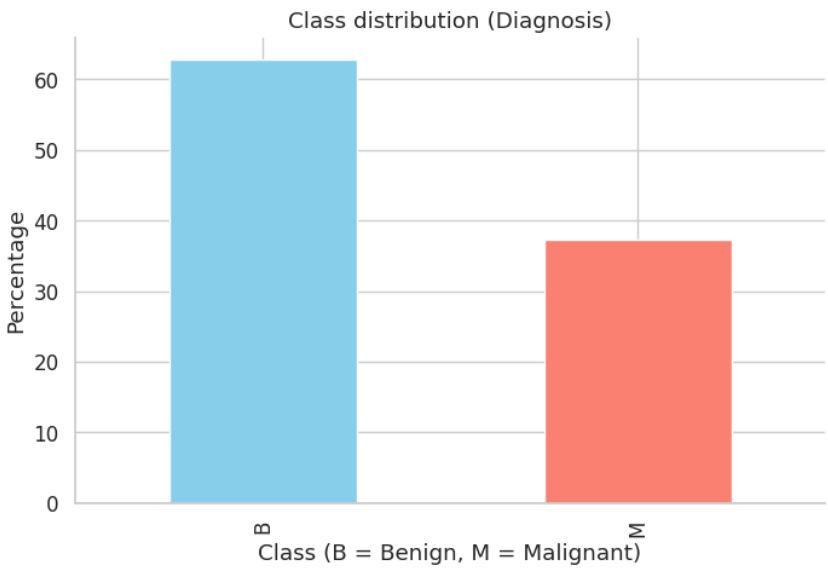


Figure 2: Class distribution

The Pearson correlation matrix of the numerical features reveals the presence of strong positive correlations among several features.

Figure 3: Pearson correlation matrix

Such strong relationships highlight a significant level of redundancy among predictors, especially within groups of geometrically related measurements (e.g., radius, concavity, and concave points across different aggregations).

To mitigate this redundancy, a feature selection step was incorporated using `SelectKBest` within the modeling pipeline. This approach allows each model to automatically retain only the most informative subset of features, reducing dimensionality and limiting the impact of multicollinearity while preserving predictive performance.

3.2 Overall Classification Performance

The classification performance of the models on the test set is summarized in Table 2.

Model	Accuracy	Precision	Recall	F1-score
Logistic Regression	0.977	0.984	0.953	0.968
SVM (RBF kernel)	0.994	1.0	0.984	0.992
LDA	0.953	1.0	0.875	0.933
KNN	0.959	0.983	0.906	0.943

Table 2: Classification performance of the models on the test set.

All models achieved high predictive performance, with accuracy values above 0.95, confirm-

ing the effectiveness of the proposed pipeline.

Among the evaluated methods, the SVM classifier achieved the best overall performance, with the highest accuracy (0.994), perfect precision (1.0), and the highest F1-score (0.992). Logistic Regression also showed strong results, with a good balance between precision and recall, although slightly lower than SVM.

KNN also achieved solid performance, although with slightly lower recall and F1-score, suggesting a higher tendency to miss some malignant cases compared to the best-performing models. LDA showed the lowest performance overall, particularly in terms of recall (0.875), indicating a reduced ability to correctly identify malignant samples.

Overall, the results highlight that SVM offers the most reliable classification performance on this dataset, especially in balancing precision and sensitivity.

3.3 ROC analysis

The discriminative ability of the models was further evaluated using Receiver Operating Characteristic (ROC) curves.

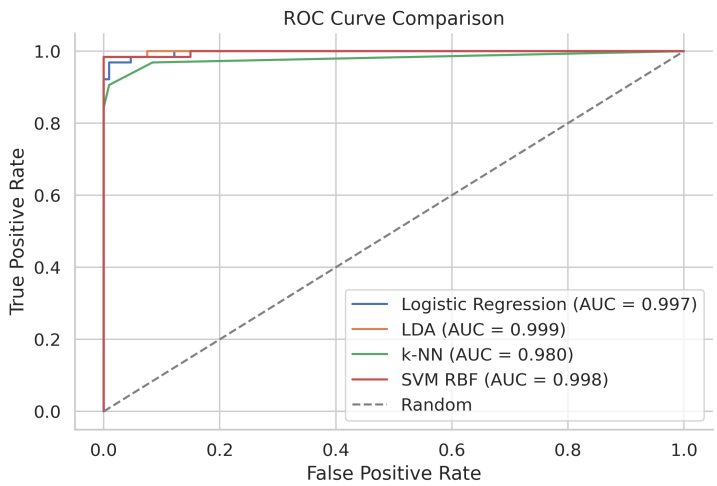


Figure 4: ROC curves for all classification models on the test set

All models achieved high area under the curve (AUC) values, close to 1, indicating an excellent ability to distinguish between benign and malignant samples. Logistic Regression reached an AUC of 0.997, while LDA achieved the highest value (AUC = 0.999). SVM also performed very well with an AUC of 0.998, whereas KNN showed slightly lower performance (AUC = 0.980).

Although the AUC values are very similar, they indicate how well the models separate the two classes, but not how accurate the final classifications are.

Overall, the ROC analysis confirms that all models are effective in separating the two classes, with LDA and SVM showing the strongest discriminative ability.

3.4 Precision–Recall analysis

Since the dataset presents a moderate class imbalance, model performance was also evaluated using Precision–Recall (PR) curves.

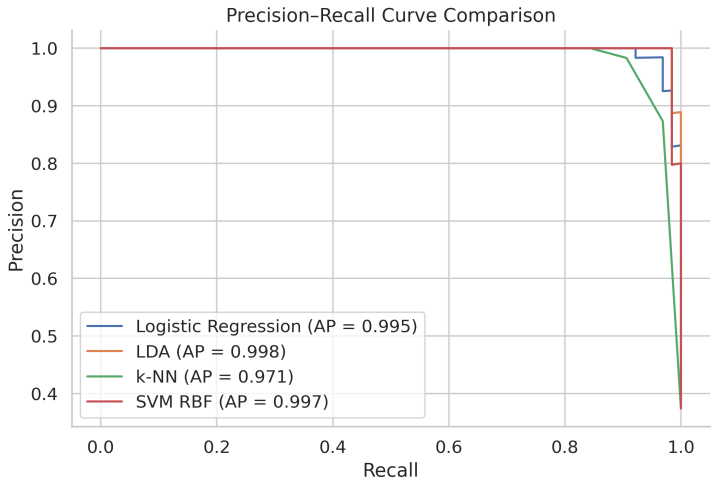


Figure 5: Precision-Recall curves for all classification models on the test set

All models achieved high precision values across a wide range of recall values, indicating good performance in identifying malignant cases.

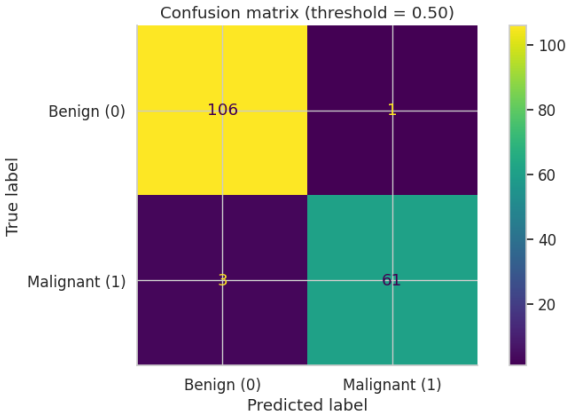
Logistic Regression reached an average precision (AP) of 0.995, while LDA achieved the highest value (AP = 0.998). SVM also performed strongly with an AP of 0.997, whereas KNN showed slightly lower performance (AP = 0.971).

Although the differences between models are relatively small, all classifiers show a drop in precision at high recall levels, reflecting the typical trade-off between detecting more positive cases and maintaining high precision.

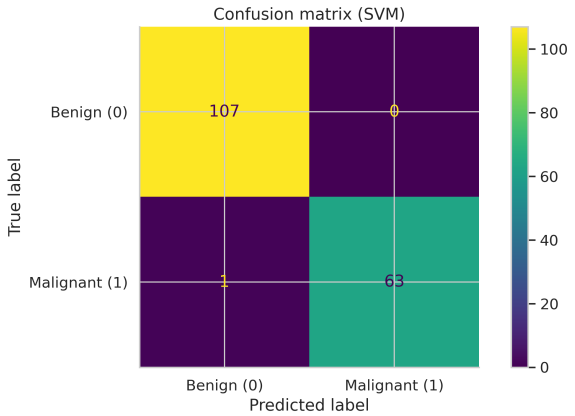
Overall, the PR analysis confirms that all models perform well on this task, with LDA, SVM, and Logistic Regression showing slightly more stable performance compared to KNN.

3.5 Error Analysis and Confusion Matrices

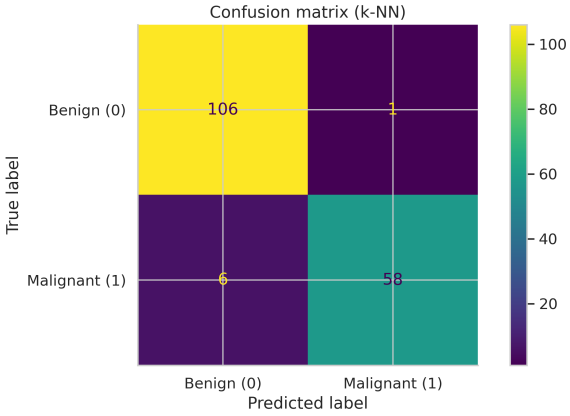
To further analyse the classification behaviour of the models, confusion matrices were computed on the test dataset.



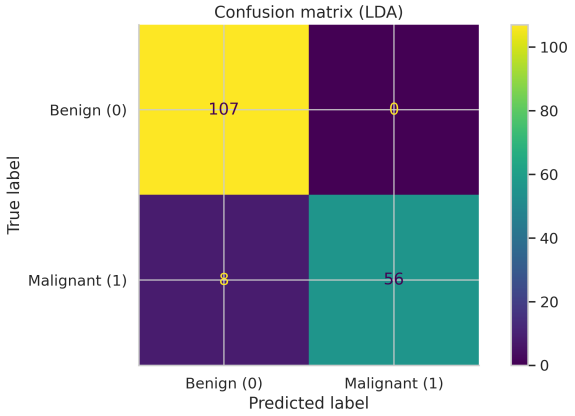
(a) Logistic Regression



(b) Support Vector Machine



(c) KNN



(d) LDA

Figure 6: Confusion matrices for all classification models on the test set.

Overall, all classifiers achieved a high number of correct predictions, with only a limited number of misclassifications.

Logistic Regression and SVM show very similar classification behaviour, both producing only a small number of errors. In particular, they correctly identify most malignant cases, with very few false negatives.

KNN presents a slightly higher number of false negatives compared to Logistic Regression and SVM, indicating a reduced ability to detect some malignant cases. LDA shows the largest number of errors among the evaluated models, mainly due to a higher number of false negatives.

These results are in line with the ROC and Precision–Recall analysis. Although LDA achieved the highest AUC and average precision, it produces more classification errors at the chosen threshold. This suggests that, despite its strong ability to separate the classes, its final predictions are more sensitive to the threshold selection.

4 Discussion

The results obtained in this study show that all evaluated models achieve very high predictive performance on the Breast Cancer Wisconsin dataset. In particular, Logistic Regression and Support Vector Machines provide consistently strong results across different evaluation metrics, confirming their effectiveness for this binary classification task.

The analysis of ROC and Precision–Recall curves further supports these findings, indicating that all models maintain high discriminative ability across different decision thresholds. This is especially relevant in a medical context, where balancing sensitivity and precision is crucial.

When focusing on classification errors, some differences emerge. SVM produces the lowest number of false negatives, followed closely by Logistic Regression, suggesting better performance in identifying malignant cases. In contrast, KNN and LDA show a higher number of false negatives, indicating lower sensitivity to the malignant class at the selected threshold.

From a clinical perspective, these results highlight the importance of minimizing false negatives, as missing malignant cases could delay diagnosis and negatively impact patient outcomes. Therefore, models such as SVM and Logistic Regression may be more suitable in settings where sensitivity is prioritized.

Despite the strong performance, some limitations should be considered. The analysis was conducted on a single dataset with a relatively limited number of samples, which may not fully capture the variability of real-world clinical data. In addition, the dataset is well-curated, with no missing values and limited noise, which may lead to optimistic performance estimates compared to more heterogeneous and noisy real-world clinical data.

5 Conclusion

This study demonstrates that classical machine learning models can achieve highly reliable performance for breast cancer classification when combined with appropriate preprocessing and model selection strategies.

Rather than identifying a single best model, the results show that multiple approaches can provide similarly strong overall performance, with differences mainly emerging in how classification errors are distributed. In particular, the ability to reduce false negatives is a key factor in model selection for medical applications.

Overall, the combination of feature standardization, feature selection, and systematic hyperparameter tuning proved effective in building a robust classification pipeline.

Future work could extend this study by including additional datasets, exploring more advanced models, and further analyzing which samples are more likely to be misclassified, in order to better understand potential sources of error.

References

Md Manjurul Ahsan, Shahana Akter Luna, and Zahed Siddique. Machine-learning-based disease diagnosis: A comprehensive review. *Healthcare*, 10(3):541, 2022. doi: 10.3390/healthcare10030541.

Xingyu Chen, Zewen Yin, and Hao Tian. Support vector machines principles and actually example. *Procedia Computer Science*, 243:2–11, 2024. doi: 10.1016/j.procs.2024.09.002.

R. K. Halder, M. N. Uddin, M. A. Uddin, et al. Enhancing k-nearest neighbor algorithm: a comprehensive review and performance analysis of modifications. *Journal of Big Data*, 11: 113, 2024. doi: 10.1186/s40537-024-00973-y.

Patrick Schober and Thomas R. Vetter. Logistic regression in medical research. *Anesthesia & Analgesia*, 132(2):365–366, 2021. doi: 10.1213/ANE.0000000000005247.

Xin Xiong, Le-Wei Zheng, Yu Ding, Yu-Fei Chen, Yu-Wen Cai, Lei-Ping Wang, Liang Huang, Cui-Cui Liu, Zhi-Ming Shao, and Ke-Da Yu. Breast cancer: pathogenesis and treatments. *Signal Transduction and Targeted Therapy*, 2025. doi: 10.1038/s41392-024-02108-4.

S. Zhao, B. Zhang, J. Yang, et al. Linear discriminant analysis. *Nature Reviews Methods Primers*, 4:70, 2024. doi: 10.1038/s43586-024-00346-y.